

Grading Report

Submission: physics_question_paper.pdf

Pages: 2

Questions: 10

31.0 / 35.0

88.6%



Question-wise Breakdown

Q1. State Newton's second law of motion and write its formula.

3.0/3.0

STUDENT ANSWER

It states that force equals rate of change of momentum. Formula: $F = m \times a$

EVALUATION

The student correctly states that force equals the rate of change of momentum, which matches the expected answer. This earns 2 marks for the statement. The formula ' $F = m \times a$ ' is also correct, earning 1 mark. Both parts of the rubric are satisfied.

RUBRIC

Newton's second law: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force. OR: Force equals mass times acceleration....

Q2. A body of mass 5 kg accelerates at 3 m/s^2 . Find the force.

3.0/3.0

STUDENT ANSWER

$F = m \times a = 5 \times 3 = 15 \text{ N}$

EVALUATION

The student correctly applied the formula ($F = m \times a$), substituted the values (5×3), and provided the correct answer with the unit (15 N). All rubric criteria are met.

RUBRIC

Correct application of $F = ma$. $F = 5 \text{ kg} \times 3 \text{ m/s}^2 = 15 \text{ N}$. Award 1 mark for formula, 1 mark for substitution, 1 mark for correct answer with unit....

Q3. Draw a free-body diagram of a block resting on a rough horizontal surface b

4.0/5.0

STUDENT ANSWER

Diagram labels: f

F

mg

EVALUATION

The diagram includes a rectangular block and four force arrows: weight (mg) downward, applied force (F) to the right, and friction (f) to the left. However, the normal force (N or R) upward is missing; instead, there is an upward arrow that is not labeled as the normal force. The diagram is otherwise neat and the arrows are labeled correctly for the forces shown.

RUBRIC

Free-body diagram of a block on a rough horizontal surface pushed by force F. Expected visual features: - Rectangular block clearly drawn...

Q4. Define kinetic energy and give its SI unit.

2.0/2.0

STUDENT ANSWER

Energy possessed by a body due to its motion. SI unit: joule (J).

EVALUATION

The student correctly defines kinetic energy as 'energy possessed by a body due to its motion,' which matches the rubric's requirement for the definition. The SI unit is correctly given as 'joule (J),' which is also acceptable per the rubric. Both points are addressed fully.

RUBRIC

Kinetic energy: The energy possessed by a body by virtue of its motion. SI unit: joule (J) or $\text{kg}\cdot\text{m}^2/\text{s}^2$. Award 1 mark for definition, 1 mark for SI unit....

Q5. Complete the motion table for uniform acceleration $a = 2 \text{ m/s}^2$, starting fr

5.0/5.0

STUDENT ANSWER

t (s) v (m/s) x (m) 0 0 0 1 2 1 2 4 4 3 6 9

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EVALUATION

The student provided the complete motion table with correct values for each row: $t=0$ ($v=0$, $s=0$), $t=1$ ($v=2$, $s=1$), $t=2$ ($v=4$, $s=4$), $t=3$ ($v=6$, $s=9$). This matches the expected answer. Although the formulas are not explicitly written, the rubric does not require the formulas to be shown if the table is fully correct. Therefore, full marks are awarded.

RUBRIC

Uniform acceleration table with $a = 2 \text{ m/s}^2$, starting from rest ($u = 0$). Using $v = u + at$ and $s = ut + \frac{1}{2}at^2$: $t=0$: $v=0$, $s=0$...

Q6. Calculate the pressure exerted by a force of 100 N on an area of 5 m^2 .

3.0/3.0

STUDENT ANSWER

$P = F / A = 100 / 5 = 20 \text{ Pa}$

EVALUATION

The student provided the correct formula ($P = F / A$), substituted the correct values ($100 / 5$), and gave the correct answer with the correct unit (20 Pa). All rubric criteria are fully met.

RUBRIC

Pressure = Force / Area. $P = 100 \text{ N} / 5 \text{ m}^2 = 20 \text{ Pa}$ (or N/m^2). Award 1 mark for formula, 1 mark for substitution, 1 mark for correct answer with unit....

Q7. Using Ohm's law, find the current when $V = 12 \text{ V}$ and $R = 3 \text{ ohm}$.

1.0/3.0

STUDENT ANSWER

$I = V \times R = 12 \times 3 = 36 \text{ A}$

EVALUATION

The student attempted to use Ohm's law but used the incorrect formula ($I = V \times R$ instead of $I = V/R$). They substituted the values and calculated 36 A, which is incorrect. According to the rubric, if the student inverts the formula and writes $I = V \times R = 36 \text{ A}$, only 1 mark is awarded for the attempt.

RUBRIC

Ohm's law: $V = IR$, therefore $I = V/R$. $I = 12 \text{ V} / 3 \text{ Ohm} = 4 \text{ A}$. Award 1 mark for correct formula, 1 mark for substitution, 1 mark for correct answer with unit....

Q8. Explain the term 'inertia' with one everyday example.**4.0/4.0****STUDENT ANSWER**

Inertia is the tendency of a body to resist a change in its state of motion. Example: passengers lurch forward when a bus stops suddenly.

EVALUATION

The student provides a correct definition of inertia ('the tendency of a body to resist a change in its state of motion'), which matches the rubric's requirement for 2 marks. The example given ('passengers lurch forward when a bus stops suddenly') is explicitly listed as a valid example in the rubric, and the explanation is clear. Both parts of the answer meet the rubric criteria.

RUBRIC

Inertia: The tendency of a body to resist any change in its state of rest or uniform motion in a straight line. Example: Any valid everyday example demonstrating inertia, e.g.:...

Q9. Write the formula for momentum and calculate it for $m = 3 \text{ kg}$, $v = 4 \text{ m/s}$.**3.0/3.0****STUDENT ANSWER**

$p = m \times v = 3 \times 4 = 12 \text{ kg.m/s}$

EVALUATION

The student provided the correct formula for momentum ($p = m \times v$), substituted the given values correctly (3×4), and calculated the correct answer with the correct unit (12 kg.m/s). All rubric criteria are fully met.

RUBRIC

Momentum formula: $p = mv$. Calculation: $p = 3 \text{ kg} \times 4 \text{ m/s} = 12 \text{ kg.m/s}$. Award 1 mark for formula, 1 mark for substitution, 1 mark for correct answer with unit....

Q10. State the law of conservation of energy.**3.0/4.0****STUDENT ANSWER**

Energy can neither be created nor destroyed, only transformed from one form to another.

- End of Physics Paper -

EVALUATION

The student correctly states that energy can neither be created nor destroyed and mentions transformation from one form to another, which covers the main points for 2 marks (correct statement) and 1 mark (mentioning transformation). However, the answer does not explicitly mention that the total energy of an isolated system remains constant, which is required for the final mark according to the rubric.

RUBRIC

Law of conservation of energy: Energy can neither be created nor destroyed; it can only be transformed from one form to another. The total energy of an isolated system remains constant....

Student Feedback

You showed a strong grasp of the basics, especially with force and motion. Your application of $F = m \times a$ in Q2 was spot on, and the motion table in Q5 was fully correct -- nice attention to detail there. The definition and example for inertia in Q8 were clear and well-chosen. For Q3, your force diagram was neat and most arrows were labeled correctly, but you missed labeling the normal force upward. Make sure to always include and label all forces acting on a body, especially the normal force in contact situations. In Q7, you tried to use Ohm's law but used the wrong formula ($I = V \times R$ instead of $I = V/R$), which led to an incorrect answer. Review the relationship between current, voltage, and resistance so you don't mix up the formula next time. For Q10, your explanation of energy conservation was good, but remember to mention that the total energy in an isolated system stays constant -- that's a key part of the principle. Solid effort overall, just tighten up your formulas and force diagrams and you'll be in great shape.

